

Enhanced control scheme for EV motor drive system with Vienna Rectifier and SVPWM

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Abstract

The electric vehicle (EV) motor drive system consists of they components, including the electric motor, power electronics, and control systems, which together convert electrical energy into mechanical power. This paper proposes a control architecture for a EV motor drive system that integrates a Vienna rectifier and Space Vector Pulse Width Modulation (SVPWM) techniques. The topology employs a 3-ph. voltage source inverter (VSI) with a Vienna rectifier to improve efficiency. To improve performance under varying load conditions, Proportional - Integral (PI) controllers are implemented for voltage and speed regulation. The switching pulses for inverter are generated by using SVPWM technique and thus maximising output voltage and minimising harmonics. The proposed system is simulated in MATLAB 2021a and the results demonstrate rapid transient response, enhanced torque control, and resilience to fluctuations. The proposed control scheme significantly advances EV drive system development and supports the adoption of environmentally sustainable transportation.

Keywords: Vienna Rectifier, Electric Vehicle (EV), Space Vector Pulse Width Modulation (SVPWM), Voltage Source Inverter (VSI), PI Controller.

1.Introduction

In the midst of global efforts to reduce carbon emissions and to enhance energy efficiency, electric vehicles (EVs) have gained prominence as viable alternative to conventional internal combustion engine vehicles. However, to optimize performance of EV and efficient power conversion, advanced control strategies are essential. This paper introduces a novel control architecture to integrate Vienna rectifier and SVPWM. It enriches the overall efficiency and stability of EV motor drive systems. The Vienna rectifier provides high power factor correction (PFC) and low harmonic distortion, while SVPWM optimizes inverter switching losses.

Over the past decade, electric propulsion systems have been advanced significantly. Multiple control methods such as Direct Torque Control (DTC), Model Predictive Control (MPC), and Field-Oriented Control (FOC), have been implemented to enhance motor efficiency, and each presenting distinct advantages and limitations. The combination of the Vienna rectifier and SVPWM proposed in this paper offers an effective alternative, characterized by reduced harmonics, increased robustness, and improved power factor correction.

2.Literature Review

Various studies have been investigated for different control techniques for EV motor drives. Among those, Model Predictive Control (MPC) and Field-Oriented Control (FOC) are widelyused for accurate torque and speed regulation. But these methods suffer from their computational complexity and can hinder real time implementation. Additionally, fuzzy logic control (FLC) and adaptive control systems have been examined for their robustness under unpredictable environmental conditions. The present research advances existing strategies by integrating a Vienna rectifier with SVPWM, achieving superior performance with reduced complexity.

Deepak Mohanraj et al [2022] identified critical aspects of electric motor drive controllers and mitigation techniques of torque ripple is vital for boosting the reliability and performance of electric vehicles (EVs). Modern control techniques such as field-oriented control (FOC) and model predictive control (MPC), significantly reduce torque ripples, causing more optimised energy utilisation and extended battery life. Moreover, effective torque ripple mitigation improves the longevity of motor components by lowering mechanical stress and vibrations, which lowers maintenance costs. However, there are notable disadvantages to be consider.

Andrea Credo *et al* [2022] proposed an adoption of synchronous reluctance motors (SRMs) in EVs, particularly with a focus on their flux weakening capability. The notable benefits of SRMs are simplicity and robustness that often lead to lower manufacturing cost. The flux weakening capability is notably advantageous, since it allows the motor to maintain performance at higher speeds, increasing the operational range of the EV.

Ranjan Muduli *et al.* [2022] developed a predictive control method with a battery power sharing scheme in EVs. This represents a considerable innovation in boosting vehicle efficiency and dynamics.

Heejune Chafort *al* [2023] developed an operational strategy for Electric Vehicle aggregators within integrated power and transportation systems. During peak load periods, EV aggregators optimise energy usage and improve grid stability by balancing the demand and supply. They facilitated the incorporation of renewable energy sources by providing financial incentives for EV owners through dynamic pricing and compensation for grid services.

Utkal Ranjan Muduli *et al* [2023] proposed a control method for dual motor power sharing in EVs with battery power management. This is a notable advancement in boosting vehicle efficiency and performance. This method offers several benefits, primarily by improving overall power distribution among the motors, leading to enhanced acceleration, traction and stability, especially under vulnerable driving conditions.

Yiyan Su *et al* [2023] presented Model Predictive Control based torque distribution method EVs. The system is used for the planar motion of induction motor in four-wheel independently driven electric vehicles (EVs). They considered motor models and iron losses, and is a considerable advancement in vehicle control strategies.

Sang Hyuk Kim *et al* [2023] presented Model Predictive Control (MPC) for energy-efficient yaw stabilising torque vectoring in electric vehicles (EVs) with four in-wheel motors. This amounts to a significant advancement in vehicle dynamics as well as control strategies.

Pemmareddy Saiteja *et al* [2023] presented the development of an energy efficient management approach to mitigate speed and torque ripples in switched reluctance motors for EVs. They implemented this through an adaptive supervisory self-learning technique. It has contributed to advancements in motor control and performance improvement.

Elie Libbos *et al* [2023] have discussed the winding layout considerations in variable pole induction motors which are critical for refining performance and efficiency. Variable pole designs were used to adjust the motor's operating characteristics based on driving conditions and enabling better torque output and efficiency at varying speeds. This feat results in more versatile power train and boosting overall performance and driving range of the vehicle.

G. Mathesh *et al* [2023] proposed a new intelligent controller based power management system with instantaneous reference currents to improve performance and energy efficiency in hybrid electric vehicles (EVs). This system adjusts power distribution among various energy sources in accordance with real time demand and operating conditions.

Paolo Pescetto *et al* [2023] developed galvanically isolated on-board chargers fully integrated with 6-phase traction motor drives, which present a significant innovation in electric vehicle (EV) technology, combining charging and propulsion systems into a cohesive unit. The incorporation of the galvanic isolation helps to protect both the vehicle and its occupants from electrical faults and reduces the risk of ground loops.

Mehdi Monadi *et al* [2024] presented that Permanent Magnet Synchronous Motors (PMSMs) are crucial in electric vehicle (EV) applications owing to their high efficiency and power density, making speed control a key aspect of optimising their performance for sustainable energy mobility. Several control techniques are employed, each with its own advantages and disadvantages. Field-Oriented Control (FOC) is a popular method, offering precise speed and torque regulation, smooth operation, and high dynamic response, but its complexity and high computational demand increase system cost.

Pemmareddy Saiteja *et al* [2024] presented adaptive self-learning-based energy management controllers for brushless DC (BLDC) motors in EVs, which are gaining attention due to their potential to optimise motor performance under real world driving conditions. These controllers leverage machine learning and adaptive algorithms to adjust parameters dynamically and hence improving efficiency and extending battery life.

Sarah Adnan Albarri *et al* [2024] introduced high gain adaptive controllers that are notably being employed in EV traction systems to enhance the performance especially under complex and dynamic driving conditions. These controllers were designed to adjust the system parameters dynamically and compensate for changes in load, speed, and environmental factors, and ensuring optimal performance.

Mahbub Ul Islam Khan *et al* [2024] presented that machine learning-driven fault detection and classification techniques are emerging as essential tools for securing electric vehicle (EV) performance by ensuring

system reliability and safety. These techniques use advanced algorithms to analyse real-time data from various EV components, such as motors, batteries, and power electronics, to detect faults before they cause significant performance degradation or system failure.

Douglas G. Scruggs *et al* [2024] developed Neural network-based cyber-threat detection strategies that are becoming increasingly relevant for enhancing the security of four motor-drive autonomous electric vehicles, which are particularly vulnerable to cyber-attacks due to their reliance on interconnected systems and autonomous functionalities

Kexiang Wei *et al* [2024] presented a research on the economic benefits and adaptability of different End-of-Life Vehicle (ELV) powertrain topologies. They highlighted the balance between cost-effectiveness, performance and flexibility of the system. Powertrain designs, including series, parallel, and hybrid configurations, offer distinct advantages.

T. Faheem Ali *et al* [2024] presented the development of a bidirectional interleaved totem pole Power Factor Correction (PFC)-based integrated on-board charger for electric vehicle (EV) Switched Reluctance Motor (SRM) drives. It extends significant advantages in terms of efficiency and integration of the system. This architecture allows the on-board charger to function both as a charger and as a power source for motor drive operations, optimising space and reducing component costs.

Md. Shahin Munsif *et al* [2024] implemented energy management systems (EMS) for EVs which are pivotal in optimising energy consumption and enhancing vehicle performance and hence extending battery life. These systems integrate various technologies, such as battery management systems (BMS), power electronics, and control algorithms, to ensure efficient energy distribution between the battery, motor, and auxiliary systems.

Juanying Zhou *et al* [2024] proposed cost based research on energy management strategies for EVs by utilising hybrid energy storage systems (HESS), which focuses on optimising the balance between performance, efficiency, and economic viability.

Despite notable advancements, current control methods, still the wide spread adoption of EVs is being rendered by high computational demands, sensitivity to parameter variations, and difficulties in real-time implementation. The proposed Vienna rectifier and SVPWM-based control architecture addresses these issues by providing high power factor, reduced harmonic distortion, and efficient torque control.

3. Proposed System

The proposed control architecture shown in Fig.3.1 comprises the following components:

- **AC Source:** Supplies power to the Vienna rectifier.
- **Vienna Rectifier:** Converts AC power to a stable DC output, ensuring improved PFC and reduced harmonic distortion.
- **PI Controller:** Regulates the output voltage and speed for optimal performance.
- **SVPWM Generator:** Generates precise PWM pulses for VSI control, maximising efficiency.
- **Three-Phase VSI:** Converts DC voltage back to AC to drive the EV motor.
- **EV Motor:** Executes propulsion based on the controlled power supply.

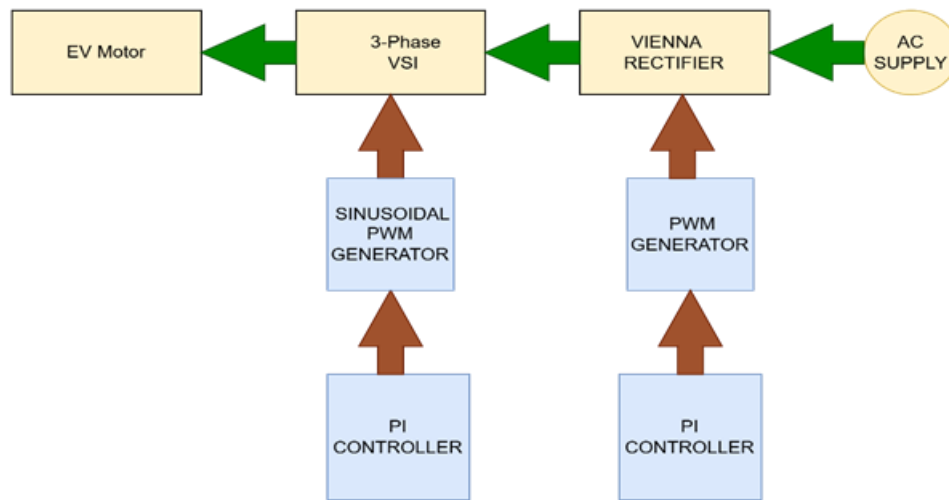


Fig.3.1- Block diagram representation of the proposed system

This paper proposes a control architecture for EV motor drive systems incorporating a Vienna rectifier and SVPWM. The system initiates with an AC source connected to the Vienna rectifier, which converts AC power into a regulated DC voltage. For optimal energy conversion, a PI controller is employed to maintain output voltage stability and to regulate power flow. The resulting DC voltage is supplied to a three-phase voltage source inverter that uses a PWM generator based on SVPWM technique for generating switching pulses. To reduce harmonic distortion, improve output voltage waveform and boosting the overall system efficiency, SVPWM technique is used. Then 3-phase VSI converts the DC voltage back into an AC which is suitable for the EV motor. A secondary PI controller regulates output voltage and current, ensuring stable operation under varying load conditions. This coordinated control architecture delivers improved dynamic response and efficiency, enabling smooth EV motor operation and maximizing battery utilization.

3.1 Operation of Vienna Rectifier

A Vienna Rectifier is a 3-level AC to DC converter with better efficiency than conventional rectifiers. It improves input power factor by reducing reactive power losses and harmonics. Figure 3.2 illustrates the functional block diagram of the Vienna rectifier within the proposed system.

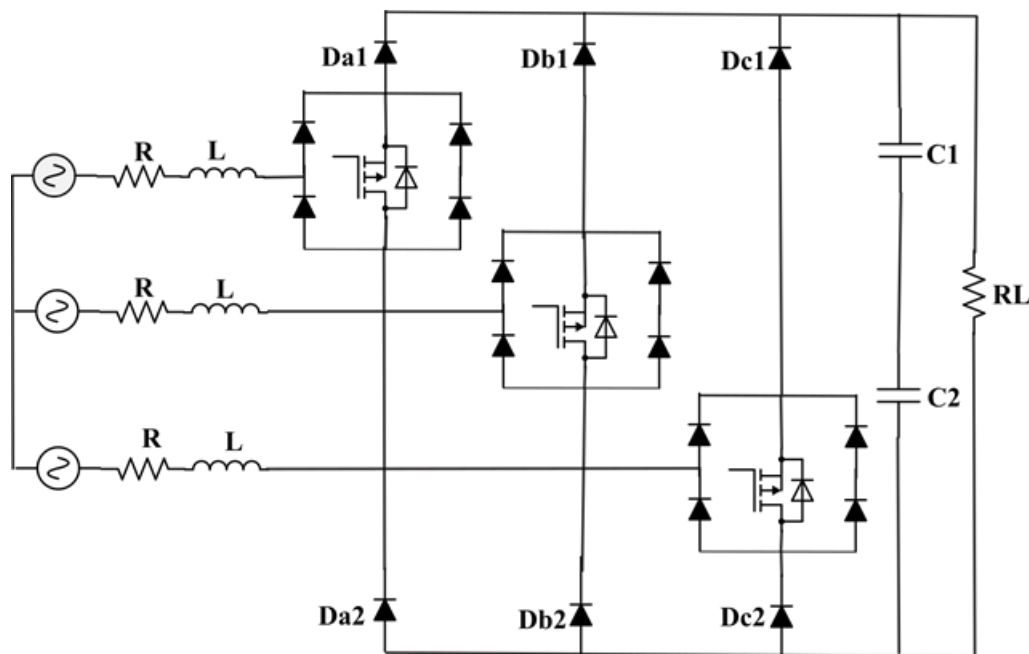


Fig.3.2- Schematic diagram of Vienna Rectifier

3.2 Implementation of SVPWM

Sinusoidal Pulse width method is an advanced modulation technique designed to enhance inverter performance. By optimizing switching states, SVPWM reduces total harmonic distortion and increases DC bus voltage utilization. The proposed system incorporates SVPWM to produce a smooth sinusoidal waveform, thereby improving motor efficiency and reducing heat generation.

4. Simulation results and Discussion

The proposed control system was simulated in MATLAB 2021a.

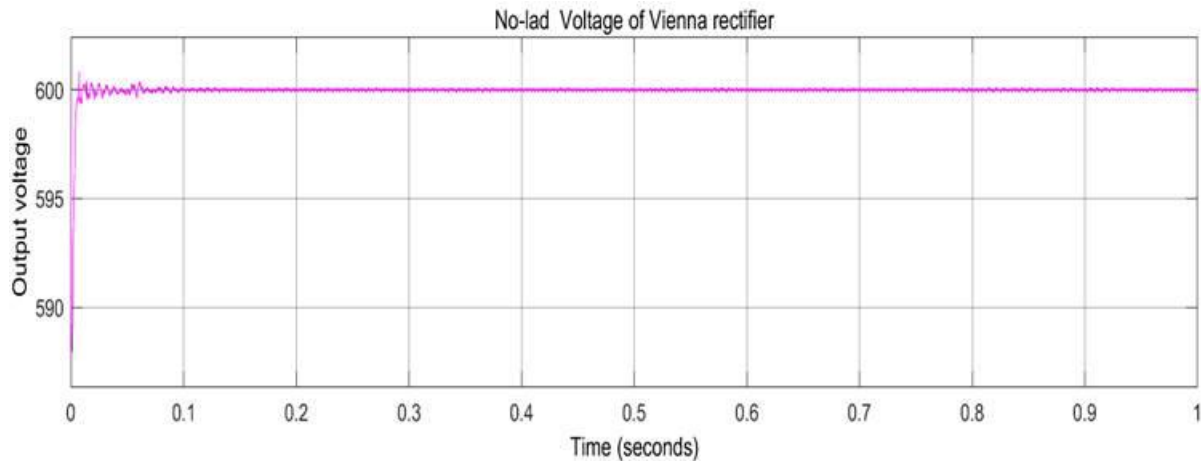


Fig 4.1 No load DC voltage of Vienna Rectifier

The waveform in Fig.4.1 exhibits no-load DC output voltage at Vienna rectifier under no load conditions. In Fig 4.2 the fluctuations in DC voltage are resulting from varying load conditions, while maintaining an overall stable trend. The reactive power waveform generally displays smaller fluctuations than real power, reflecting its function in maintaining voltage levels and supporting the operation of electrical devices.

The DC link voltage and power factor at the Vienna rectifier are illustrated in Fig.4

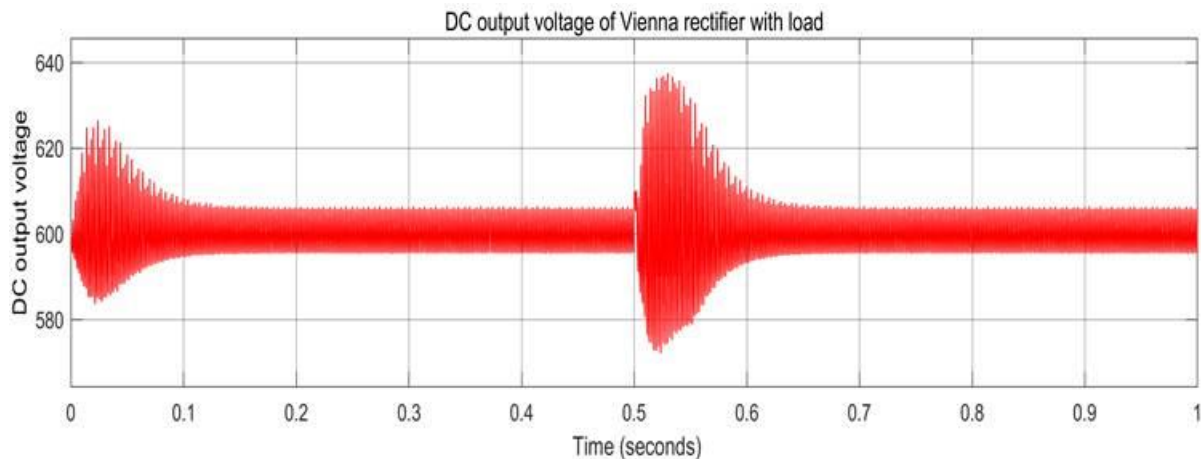


Fig. 4.2 DC output voltage of Vienna Rectifier

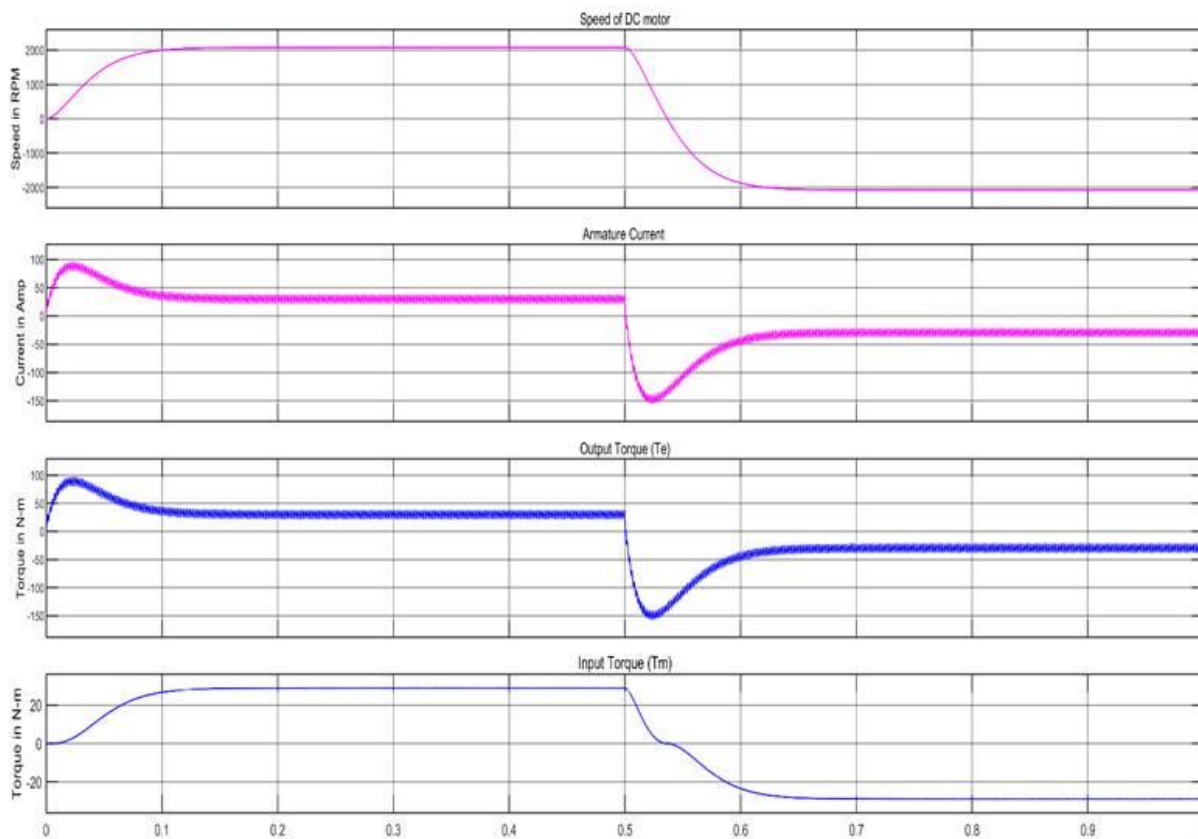


Fig. 4.3 (a) Speed of DC motor (b) Armature current (c) Electrical output torque (d) Mechanical input T_m

Figure 4.3 illustrates the performance of a DC motor with a load of 30 N-m. The upper graph depicts the speed waveform, indicating the acceleration to approximately 2000 RPM in less than 0.1 sec, followed by steady-state operation. The lower graph displays the armature current waveform, which initially shows a spike before stabilizing near 27 A.

The results demonstrate:

- **Improved Power Factor:** The Vienna rectifier achieved a power factor close to unity.
- **Reduced Harmonic Distortion:** SVPWM minimised Total Harmonic Distortion (THD) in the inverter output.
- **Better Torque and Speed Control:** The PI controller ensured smooth acceleration and decelerations

5. Conclusion

The proposed control architecture with integrated Vienna rectifier and SVPWM provides a robust and efficient solution for EV motor drive systems. Simulation results confirm its benefits in improving power quality, reducing harmonics, and thus enhancing motor performance. Future research should focus on real time prototype implementation and further optimization of control algorithms. Additionally, the application of AI driven control techniques may improve system adaptability to varying load conditions.

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